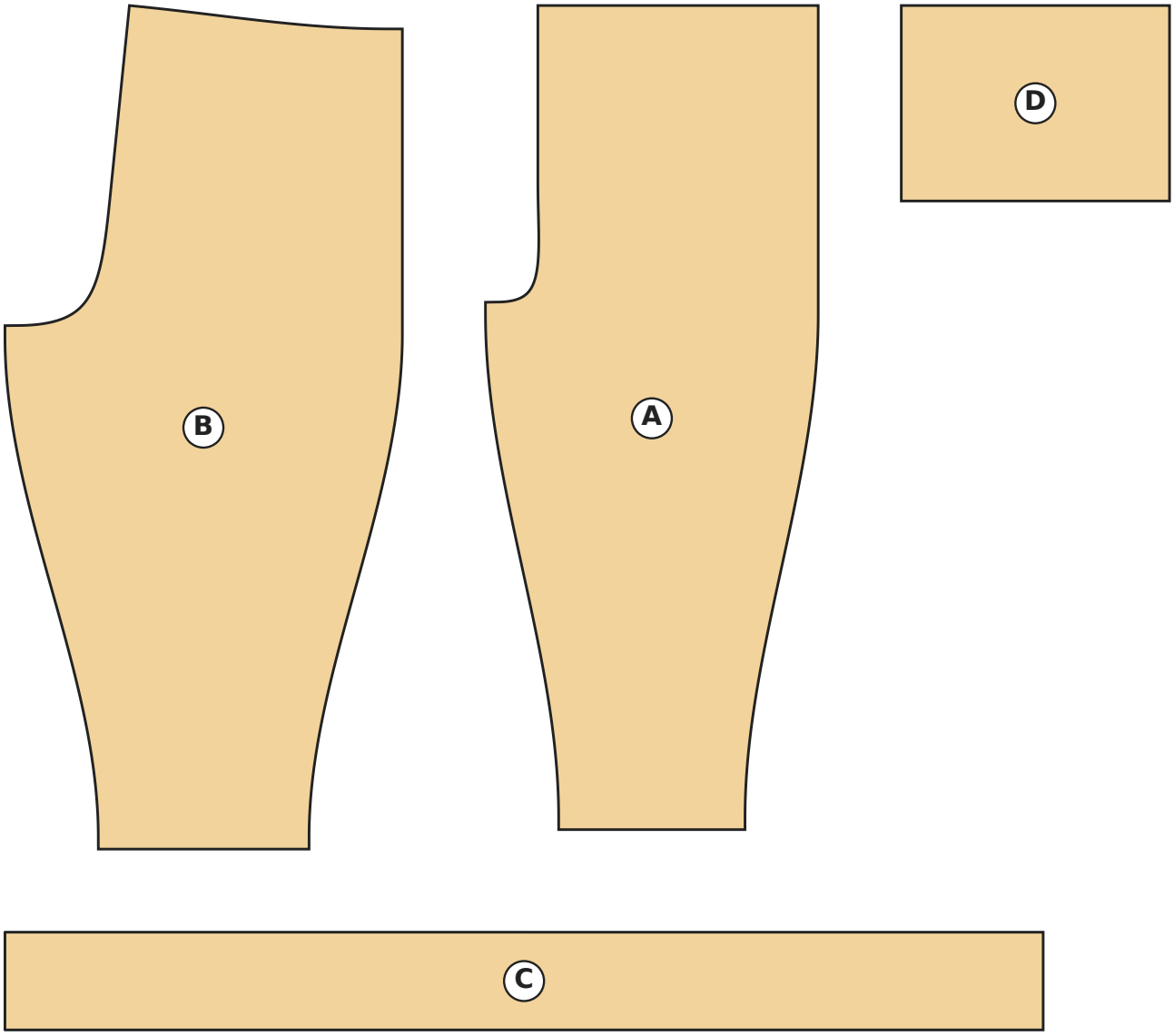
# HAREM PANTS

size 122

harem (wide, dropped crotch) fit, Sweat / French Terry (25 % stretch)




**4 parts · 20 steps**

Read the steps once before you start. Every picture is drawn from your pattern pieces.

right side · wrong side · stitch · pin · notch: match

# Parts



**A** 2×
Front leg



**B** 2×
Back leg



**C** 1×
Waistband (elastic)
cut on fold



**D** 2×
Leg cuff

# Fabric and notions

- Main fabric: 0.90 m at 150 cm width
- Rib knit: 0.25 m at 70 cm width
- Elastic 20 mm wide, length 50 cm (+2 cm overlap for joining)
- Polyester sewing thread to match

# Machines and tools



**Sewing machine**
narrow zigzag 1 × 2.5 mm (or overlocker); wide zigzag 4 × 2.5 mm; twin needle 4 mm, stitch 3 mm



**Iron**
steam



**Pins or clips**



**Fabric scissors**



**Tape measure**



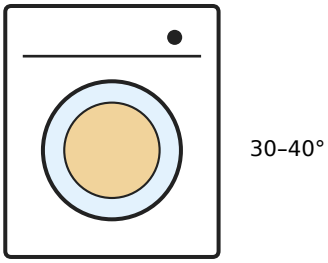
**Chalk or a fabric pen**



**Safety pin**

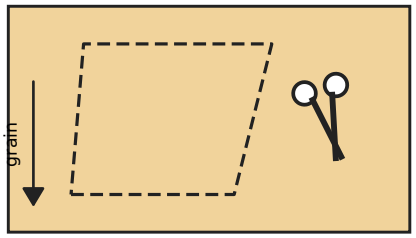
Steps are written for a sewing machine with zigzag. Have an overlocker or a coverstitch machine? Tell the generator which machines you own and the steps use them.

## 1 PREPARE
### Pre-wash the fabric



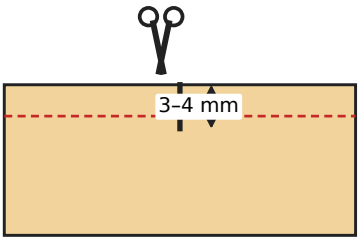

Wash and dry it the way you will wash the garment later, so it shrinks now, not after sewing.

## 2
### Cut out all pieces



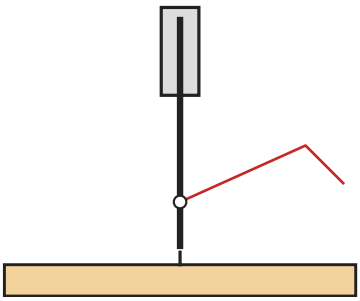

Follow the cutting layout; grainline arrows parallel to the selvage.

## 3
### Snip the notches



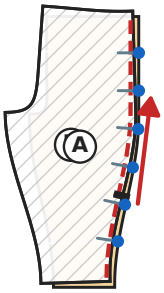

Cut 3–4 mm into the seam allowance at every notch.

## 4
### Set up the machine



Jersey/stretch needle 75–80, polyester thread. Sew a test seam on a scrap first.

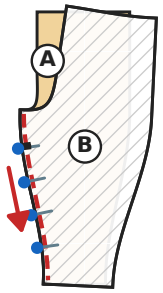
## 5 LEGS
### Sew the side seams
×2




Front and back leg right sides together; single notches = front, double = back.

Sewing machine: narrow zigzag 1 × 2.5 mm (or overlocker)

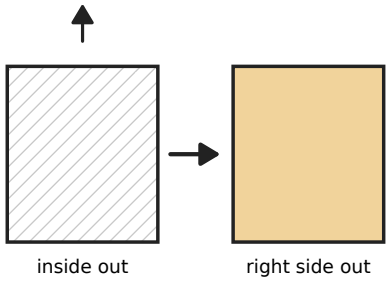
## 6
### Sew the inseams
×2




The notches on the inseam meet.

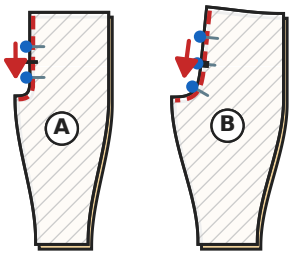
Sewing machine: narrow zigzag 1 × 2.5 mm (or overlocker)

## 7 CROTCH
### Put one leg inside the other




Turn one leg right side out and slip it into the other (wrong side out): the legs lie right sides together.

## 8
### Sew the crotch seam

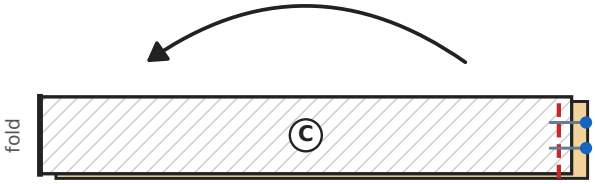



**! Sew the crotch curve a second time for strength.**

In one go from the front to the back top edge; the inseams meet exactly.

Sewing machine: narrow zigzag 1 × 2.5 mm (or overlocker)

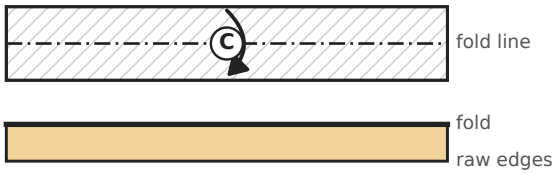
## 9 WAISTBAND
### Close the waistband into a ring




Right sides together across the short ends (centre back); leave 3 cm open for the elastic.

Sewing machine: narrow zigzag 1 × 2.5 mm (or overlocker)

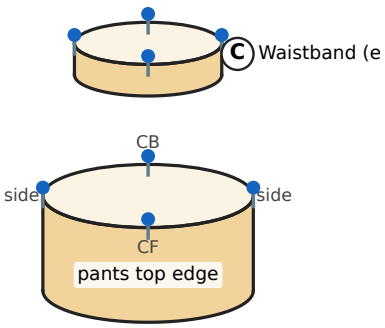
## 10
### Fold the waistband in half




Lengthwise, wrong sides together, along the fold line.

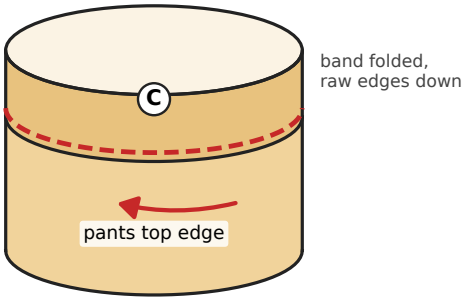
Iron: steam

## 11
### Mark quarters




On waistband and pants top edge: centre front, centre back, sides.

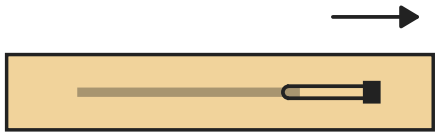
## 12
### Sew the waistband on




Raw edges together, opening at the back on the inside; quarter marks meet.

Sewing machine: narrow zigzag 1 × 2.5 mm (or overlocker)

## 13 Thread the elastic

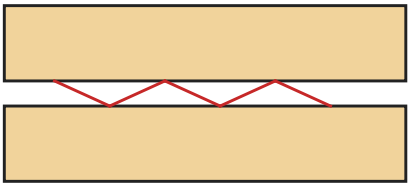


elastic, with a safety pin

With a safety pin through the opening; overlap the ends by 2 cm and sew them together.

Sewing machine: wide zigzag 4 × 2.5 mm

## 14 Close the opening



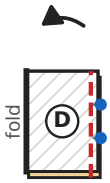
ladder stitch: the stitches disappear

Spread the gathers evenly, close the opening and stitch in the ditch at the side seams so the elastic cannot twist.

Sewing machine: twin needle 4 mm, stitch 3 mm

## 15 Close the cuffs into rings ×2

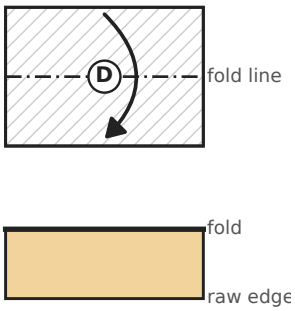
HEMS




Right sides together across the short ends.

Sewing machine: narrow zigzag 1 × 2.5 mm (or overlocker)

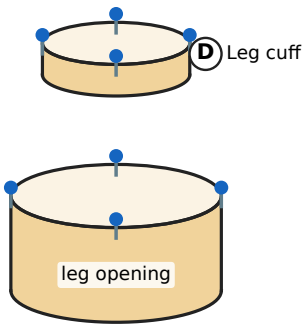
## 16 Fold the cuffs in half ×2




Lengthwise, wrong sides together.

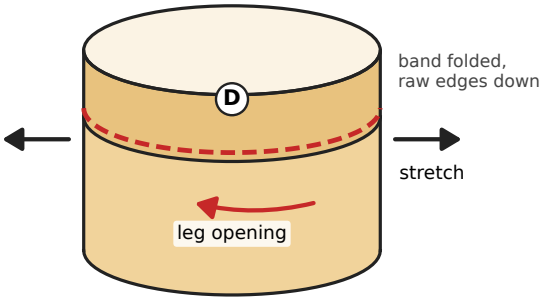
Iron: steam

## 17 Mark quarters ×2




On cuff and leg opening.

## 18 Sew the cuffs on ×2

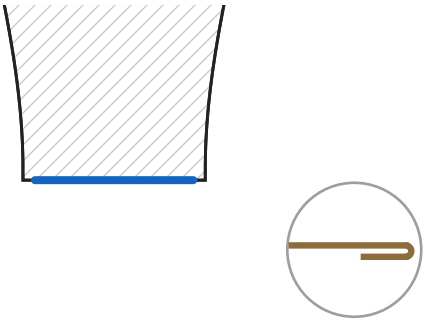



Slip the cuff over the right-side-out leg, cuff seam at the inseam.

Sewing machine: narrow zigzag 1 × 2.5 mm (or overlocker)

## 19 Press the allowance up

×2



Seam allowance towards the pants.

Iron: steam

## 20 FINISH Done!



Press all seams with steam; do not press down hard on sweat.

# Harem pants · size 122

harem (wide, dropped crotch) fit, Sweat / French Terry (25 % stretch) · original parametric draft. The pattern is yours to use.

> **Important:** With a new pattern, first sew a **test garment** from inexpensive fabric with similar stretch and check the fit.

## Files

- PDF A4 tiles to glue (12 pages + overview): `harem_pants_size122_a4_tiled.pdf`
- Preview A4 tiles: `harem_pants_size122_a4_preview.png`
- SVG master 1:1: `harem_pants_size122_master_1to1.svg`
- Cutting layout: `cutting_layout.png`
- Sewing instructions (PDF): `sewing_instructions.pdf`
- Sewing instructions (Markdown): `sewing_instructions.md`

## Printing

- Always print at **100 % / "Actual size"** (not "Fit to page").
- Measure the **5 × 5 cm** control square before you cut.
- Plotter (610 mm roll): the page is 600 mm wide in portrait. In the print dialog choose a custom paper size with exactly this width and length, and turn rotation/scaling off.

## Body measurements

| Body measurement | Value | Source |
|---|---|---|
| Body height | 122.0 cm | table |
| Chest | 62.0 cm | table |
| Waist | 56.0 cm | table |
| Hip | 66.0 cm | table |
| Head | 52.0 cm | table |
| Neck | 28.5 cm | table |
| Inseam | 54.0 cm | table |
| Side length | 72.0 cm | table |
| Arm length | 47.0 cm | table |
| Back length | 29.5 cm | table |
| Shoulder width | 10.0 cm | table |
| Thigh | 37.0 cm | table |
| Ankle | 17.8 cm | table |

## Finished measurements (virtually sewn)

| Finished measurement | Value |
|---|---|
| Hip width | 85.8 cm |
| Thigh width | 55.5 cm |
| Leg width at mid-leg | 42.0 cm |
| Leg opening | 20.0 cm |
| Waistband relaxed | 50.4 cm |
| Waistband stretched (pants top edge) | 83.0 cm |
| Outseam | 75.5 cm |
| Inseam | 48.2 cm |
| Front rise | 27.3 cm |
| Back rise | 29.1 cm |
| Waistband height | 3.0 cm |
| Cuff height | 7.0 cm |

| Finished measurement | Value |
|---|---|
| Crotch seam total | 64.2 cm |

## Pattern pieces

| Piece | Cutting | Fabric | Seam allowance |
|---|---|---|---|
| Front leg | cut 2 (mirrored pair) | Main fabric | seam allowance 1.0 cm included |
| Back leg | cut 2 (mirrored pair) | Main fabric | seam allowance 1.0 cm included |
| Waistband (elastic) | cut 1 on fold | Main fabric | seam allowance 1.0 cm included |
| Leg cuff | cut 2 | Rib knit | seam allowance 1.0 cm included |

## Fabric needed

- **Main fabric:** about 0.90 m at 150 cm width (folded selvage to selvage, cut on fold pieces at the fold, incl. 10 % for shrinkage; single layer would need 0.90 m)
- **Rib knit:** about 0.25 m at 70 cm width (single layer, incl. 10 % for shrinkage)
- Elastic 20 mm wide, length 50 cm (+2 cm overlap for joining)

Cutting layout: see `cutting_layout.png` (grainline arrow parallel to the selvage, "flipped" = place the piece face down, "×2" = cut through both layers of the folded fabric, which gives a mirrored pair, "(unfolded)" = cut-on-fold piece shown opened out).

**Cut on fold** (Waistband (elastic)): the paper piece is only one half. Its fold edge (dash-dot line, bracket arrows "PLACE ON FOLD") is the centre of the piece: it gets no seam allowance and is never cut. On the folded fabric, lay the fold edge exactly on the fabric fold, pin it and cut through both layers; opened out you get the whole piece. Where the cutting layout shows a piece "(unfolded)" on a single layer, fold the fabric along the grain just wide enough for the piece and place the fold edge on that fold, or trace the paper piece, flip it over along the fold edge and trace the second half.

## Validation report

### Sewability check

| Area | Check | Target | Actual | Δ | Result | Note |
|---|---|---|---|---|---|---|
| Seam pairs | Side seam front ↔ back | 66.0 cm | 66.0 cm | +0.0 mm | □ OK | |
| Seam pairs | Inseam front ↔ back | 41.7 cm | 41.7 cm | +0.0 mm | □ OK | |
| Seam pairs | Front crotch seam (left ↔ right piece) | 26.9 cm | 26.9 cm | +0.0 mm | □ OK | |
| Seam pairs | Back crotch seam (left ↔ right piece) | 31.3 cm | 31.3 cm | +0.0 mm | □ OK | |
| Seam pairs | Waistband ↔ pants top edge | 83.0 cm | 83.0 cm | +0.0 mm | □ OK | |
| Seam pairs | Leg cuff ↔ leg opening (stretched) | 70–85 % of 28.5 cm | 20.0 cm = 70 % | -85.5 mm | □ OK | rib band is stretched while sewing |
| Angles/transitions | Centre front ↔ waistband | 90° | 90.0° | +0.0° | □ OK | |
| Angles/transitions | Centre back ↔ waistband | 90° | 90.0° | -0.0° | □ OK | |
| Angles/transitions | Crotch ↔ inseam (front + back) | 180° | 180.9° | +0.9° | □ OK | |
| Angles/transitions | Inseam ↔ hem front | 90° | 90.1° | +0.1° | □ OK | |
| Angles/transitions | Hem ↔ side seam front | 90° | 90.1° | +0.1° | □ OK | |
| Angles/transitions | Inseam ↔ hem back | 90° | 90.2° | +0.2° | □ OK | |
| Angles/transitions | Hem ↔ side seam back | 90° | 90.2° | +0.2° | □ OK | |

| Area | Check | Target | Actual | Δ | Result | Note |
|---|---|---|---|---|---|---|
| Angles/transitions | Side seam ↔ waistband back | 90° | 90.1° | +0.1° | □ OK | |
| Angles/transitions | Side seam ↔ waistband (front + back) | 180° | 180.1° | +0.1° | □ OK | |
| Measurements | Hip circumference pattern = body + ease | 85.8 cm | 85.8 cm | +0.0 % | □ OK | body 66.0 cm + +30 % ease |
| Measurements | Top edge fits over the hip | ≥ 67.3 cm | 95.4 cm | +281.1 mm | □ OK | top edge incl. usable stretch ≥ hip |
| Measurements | Front rise plausible | 14.6 cm – 31.4 cm | 27.3 cm | +0.0 mm | □ OK | body rise 18.0 cm, harem drop 7.3 cm |
| Measurements | Inseam length plausible | 39.0 cm – 60.5 cm | 55.5 cm | +0.0 mm | □ OK | body inseam 54.0 cm |
| Measurements | Foot fits through the leg opening | ≥ 24.1 cm | 29.5 cm | +54.8 mm | □ OK | heel-instep circumference ≈ 1.35 × ankle |
| Geometry | Front leg: outline & seam allowance | closed, SA present | ok | | □ OK | seam allowance 1.0 cm included |
| Labels | Front leg: grainline & cutting | present | cut 2 (mirrored pair) | | □ OK | Main fabric |
| Geometry | Back leg: outline & seam allowance | closed, SA present | ok | | □ OK | seam allowance 1.0 cm included |
| Labels | Back leg: grainline & cutting | present | cut 2 (mirrored pair) | | □ OK | Main fabric |
| Geometry | Waistband (elastic): outline & seam allowance | closed, SA present | ok | | □ OK | seam allowance 1.0 cm included |
| Labels | Waistband (elastic): grainline & cutting | present | cut 1 on fold | | □ OK | Main fabric |
| Geometry | Leg cuff: outline & seam allowance | closed, SA present | ok | | □ OK | seam allowance 1.0 cm included |
| Labels | Leg cuff: grainline & cutting | present | cut 2 | | □ OK | Rib knit |
| Geometry | Waistband (elastic): cut-on-fold edge | straight, no SA, 90° corners, on grain | ok, corners ≤ 0.0° off 90° | | □ OK | Centre front |
| Labels | Front leg: notches | on the cutting line | 3 notch mark(s) | | □ OK | |
| Labels | Back leg: notches | on the cutting line | 6 notch mark(s) | | □ OK | |

**Result:** 31 OK, 0 warning(s), 0 error(s): the pattern is sewable.

## Construction values

Override any of these with the option `construction`, values in mm (e.g. `{"construction": {"hip_width": 858}}`).

| Name | Value | Derived from | Note |
|---|---|---|---|
| `hip_width` | 85.8 cm | hip × (1 + 30 % ease) | |
| `front_back_offset` | 0.5 cm | front quarter narrower, back quarter wider by this amount | |
| `front_width` | 20.9 cm | hip_width / 4 − front_back_offset | |
| `back_width` | 21.9 cm | hip_width / 4 + front_back_offset | |
| `front_crotch_extension` | 4.3 cm | 5 % of hip_width | |

| Name | Value | Derived from | Note |
|---|---|---|---|
| `back_crotch_extension` | 8.6 cm | 10 % of hip_width | |
| `rise` | 24.3 cm | (side length − inseam) + seat ease + harem drop − waistband height | |
| `inseam_finished` | 48.2 cm | inseam − allowance to the floor − harem drop + blousing | |
| `hem_width` | 28.5 cm | cuff length / cuff stretch ratio | |

Cutting layout · Harem pants size 122

